Review: Equations of lines

1

a Any vertical line has an undefined slope but still has an equation.

b The slope is incorrect. The slope of $7x + y = 3$ is -7.

2

a $y = 18x - 12$

b Slope: 18, y-intercept: -12

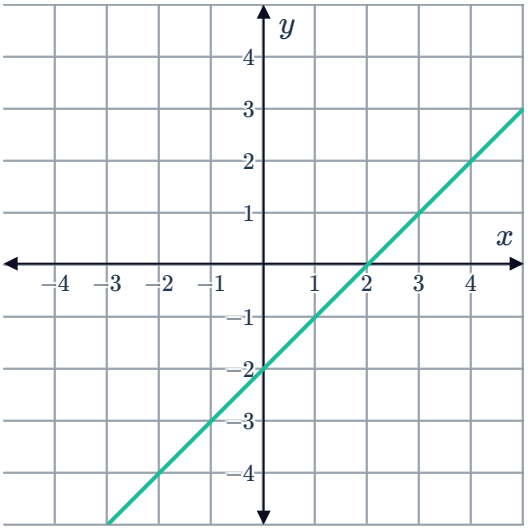
3

a $y = -8x - 9$

b $y = -7x + 7$

c $y = -2x - 13$

4



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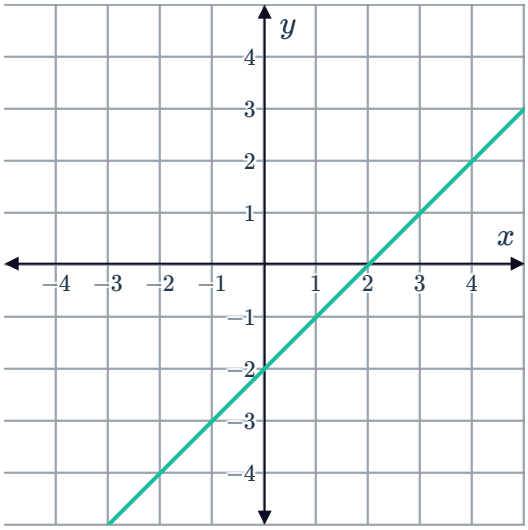
5

a $y = -2$

b $x = 2$

c $y = -7$

d



6

a $5x - y = -5$

b $2x - y = 5$

c $2x - 5y = -20$

7

a $b = \frac{40}{3}$

b $14x - 3y = -40$

8

a No

b Yes

c No

d No

9

a No

b Yes

c No

d Yes

10

a $-\frac{7}{10}$

b $y = -\frac{7x}{10} - \frac{29}{10}$ or $7x + 10y = -29$

11 $3x - 35y = -11$

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12 $5x + 8y = 40$

13

a $x - 14y = 203$

b $x = -7$

c $y = -15$

Additional questions

14

a

i $m = 6$

ii $b = 4$

b

i $m = -3$

ii $b = -8$

15

a

i $m = 3$

ii $b = -4$

iii $y = 3x - 4$

b

i $m = -\frac{3}{2}$

ii $b = 3$

iii $y = -\frac{3}{2}x + 3$

16 Line 1: $y = 2 + 6x$
Line 3: $y = 6x$
Line 6: $y = 6x - 2$

17

a Yes, $(0, 2)$ **b** Yes, $(0, 0)$ **c** Yes, $(0, 1)$ **d** No

18

a $y = -3x + 1$

b $m = -3$

c $y = 1$

19

a Yes **b** No **c** Yes **d** No

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20

a $y = -4x - 3$

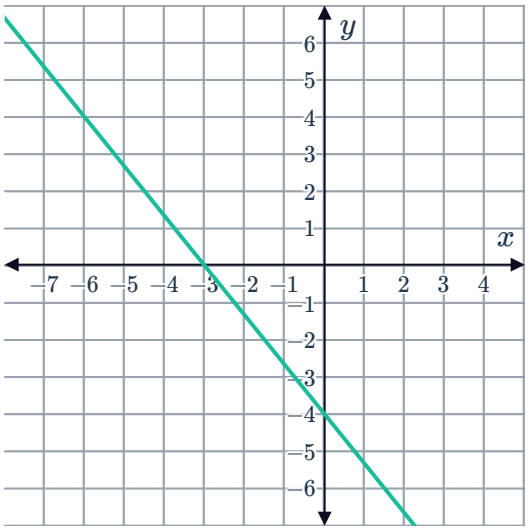
b $y = \frac{2x}{3} + 5$

c $y = -2$

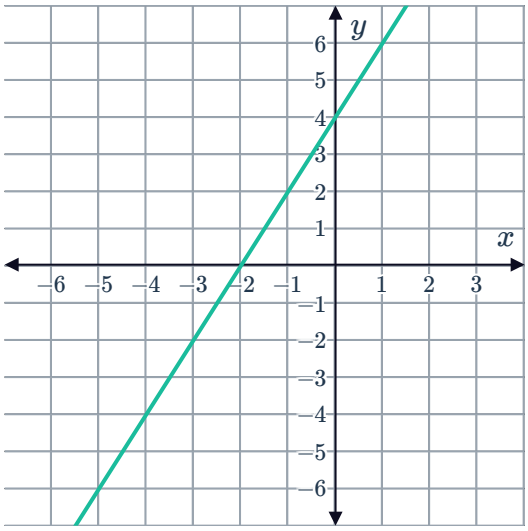
d $x = 11$

21

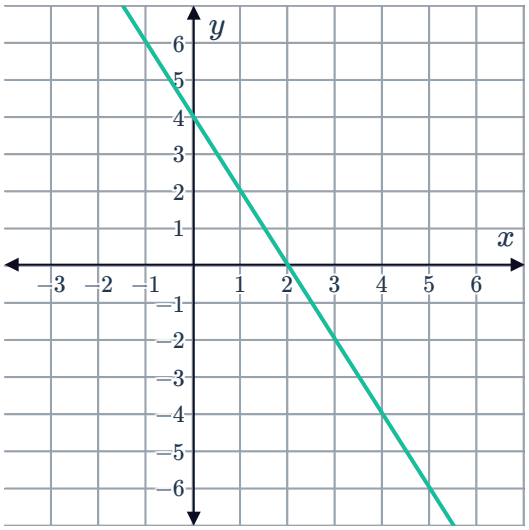
a



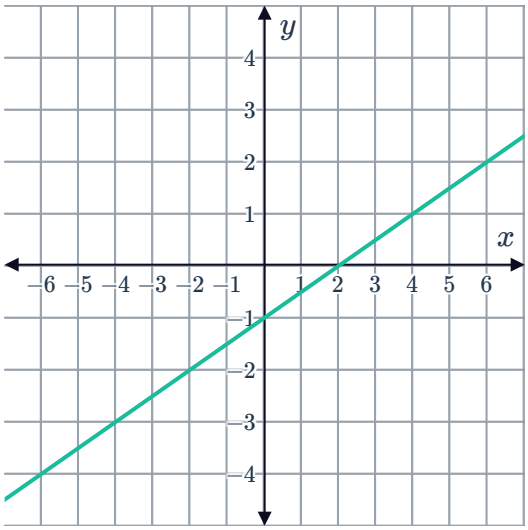
b



c

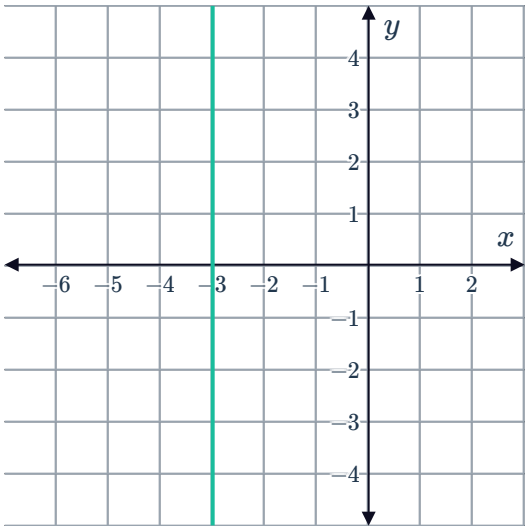


d



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e



22

a

x	-1	0	1	2
y	6	2	-2	-6

b $m = -4, b = 2$

c $y = -4x + 2$

23

a $m = \frac{1}{4}, b = -1$

b $y = \frac{x}{4} - 1$

24

a

i $m = 2$

ii $y = 2x + 5$

b

i $m = -1$

ii $y = -x + 6$

c

i $m = -\frac{2}{3}$

ii $y = -\frac{2}{3}x + 4$

d

i Undefined

ii $x = -7$

25 $Ax + By = C$, where A, B, and C are integers, and both A and B are not 0.

26

a Standard form

b Standard form

c Not in standard form

d Standard form

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27 **a** 2 **b** -3

c $y = -3x + 2$ **d** $3x + y = 2$

28 For slope m and point (x_1, y_1), point-slope form: $(y - y_1) = m(x - x_1)$

29

a Slope-intercept form **b** Point-slope form

c Standard form

30

a $y - 1 = 4\,(x - 7)$ **b** $y = -5\,(x - 4)$

c $y - 6 = 0$

31

a

i 3 **ii** $y - 6 = 3\,(x - 7)$

b

i -4 **ii** $y + 8 = -4\,(x - 4)$

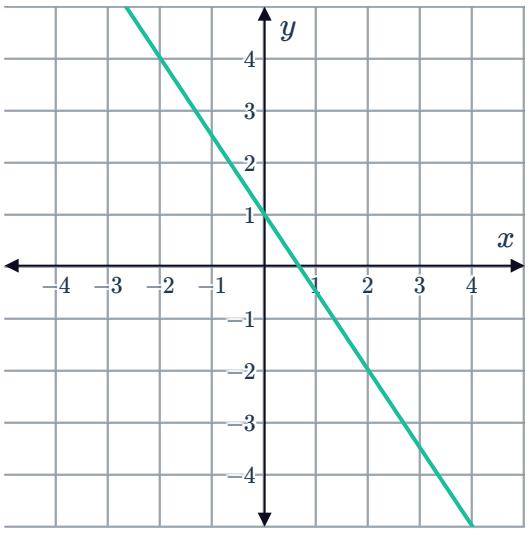
32

a $b = 1$

b $y = -\dfrac{3x}{2} + 1$

c $y = -\dfrac{3x}{2} + 1$

d



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